

Luis Martinez

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EDUCATION

University of Texas - Rio Grande Valley

Bachelor of Science in Computer Science

Edinburg, TX

Aug. 2024 – May 2026

South Texas College

Associate's in Computer Science

McAllen, TX

Aug. 2020 – May 2023

EXPERIENCE

Avionics Lead and Software Engineer

Rocket Launchers at UTRGV

September 2024 – Present

Edinburg, TX

- Led avionics development for L3 rocket, overseeing firmware, hardware, and software divisions, including PCB schematics, custom driver development, and sensor integration.
- Conducted HIL/SIL testing and collaborated with flight dynamics subteam to ensure compliance with IREC guidelines and optimize telemetry performance.

IT Analyst

Sky IT Group

November 2022 – May 2024

Weslaco, TX

- Resolved hardware/software issues and optimized networks with Ubiquiti WISP, VPNs, and automated MSSQL backups.

PROJECTS

MARV (Modular Avionics for Rockets and Vehicles) | Rust, RTIC, rp-hal, PID Control, Sensor Fusion, RP2350

- Designed modular flight computer for drone (ESC) and rocket (servo controller), implementing GNC framework with PID control and sensor fusion for real-time stability.
- Developed custom drivers, conducted environmental testing, and integrated dual-core architecture for dynamic control tasks.

L3 Rocket Air Brakes and Simulation | Python, RocketPy, Kivy, RocketSerializer

- Coordinated air brake development with SolidWorks designs and CFD analysis, integrating RocketPy simulations with Kivy UI for trajectory optimization.
- Utilized RocketSerializer to streamline data transfer from OpenRocket, enhancing simulation accuracy for control system design.

Brunito Avionics System | STM32, RTOS, DMA, Sensors (BMI088, BMM150, BMP280, u-blox GPS)

- Engineered flight computer with bidirectional telemetry and velocity-triggered payload release for IREC, optimizing firmware to enable in-flight data logging.
- Performed vacuum testing and servo calibration for reliable operation under flight conditions.

SENTINEL Telemetry GUI | Rust, Tauri, React, Real-Time Data Visualization

- Developed system for 3D orientation tracking and trajectory visualization, integrating live sensor data from avionics hardware.

ORGANIZATIONS

Rocket Launchers at UTRGV

Edinburg, TX

Avionics Lead

- Demonstrated leadership by guiding interdisciplinary teams, fostering collaboration, and adapting to project challenges under tight deadlines.
- Enhanced problem-solving and communication skills through coordinating subteams and providing technical guidance.

Society of Aerospace and Robotic Engineers (SARE)

Edinburg, TX

Project Lead

- Spearheaded end-to-end project development, demonstrating initiative in hardware design, embedded programming, and software integration to deliver a complete product.

TECHNICAL SKILLS

Languages: Rust, C++, Python, JavaScript, TypeScript

Frameworks/Tools: RTIC, rp-hal, RocketPy, Kivy, RocketSerializer, Tauri, React, DMA, PID Control, Sensor Fusion

Aerospace Skills: Embedded Firmware, PCB Design, Datasheet Analysis, HIL/SIL Testing, Telemetry Systems, LoRa Communication, CFD Integration